without a trace and without harm to the environment.

Scientists have already designed transient electronics in the form of temperature sensors, solar cells and miniature digital cameras, for instance. However, earlier versions of bio-resorbable devices were made of materials that only partially dissolved, leaving behind residues, and did not perform as well as current devices. Though this area is at present largely unexplored, researchers are emphasising on the potential applications of transient electronics in the commercial world. Given below are some exotic applications of transient electronics.

Scientists have made key advances towards practical uses of a new genre of tiny, biocompatible electronic devices that could be implanted into the body to relieve pain or battle infection for a specific period of time and then dissolve harmlessly. The medical device, once its job is done, could harmlessly melt away inside the body.

Researchers are conducting further studies, centred around degradable polymer based materials that would make suitable platforms for other electronic components, including work on transient LED transistor technology. They have produced a blue LED mounted on a polymer base with electrical leads embedded on it. When it comes into contact with a drop of water, the base and leads begin to dissolve and the light goes out.

A lost credit card could vanish from existence (but most likely still leave debt behind), a secret diary could be programmed to self-destruct, should it be removed from its hiding spot, and sensors stored with food could indicate when it has reached temperatures that would cause the food to spoil.

The real-world application for transient electronics is, perhaps, in the field of military espionage. Should a spy or informer carrying sensitive information be captured, injured or worse, the electronics could be triggered to melt away before any classified information is gleaned by enemy forces.

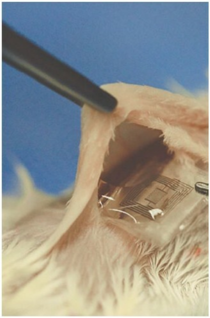

Fig. 2: This electronic implant can dissolve inside the body of a mouse (Image courtesy: http://rational-trader.blogspot.in)

A military device could collect and send data and then dissolve away, leaving no trace of an intelligence mission.

An environmental sensor could collect climate information and then wash away in the rain.

Electronic waste can also be controlled by designing integrated circuits out of materials that are biodegradable. Transient electronics, the new class of silk-silicon devices, promises a generation of medical implants that would never need surgical removal, as well as environmental monitors and consumer electronics that can become compost rather than trash.

Developmental status

Transient electronics can go as far and as complex as the demand and applications allow. Development of self-destructing devices came from a broad multi-disciplinary collaboration uniting researchers from all fields of science and technology. Multi-disciplinary research groups have tackled the problem of using other triggers to break down devices, including ultraviolet light, heat and mechanical stress. The goal is to find ways to disintegrate the devices so that manufacturers can recycle costly materials from used or obsolete devices, or so that devices could break down in a landfill.

Previous research in the area has explored the use of transient materials to create dissolvable devices such as transistors, resistors and diodes. Describing the research, researchers have mentioned that polymer composites consist of different ratios of gelatin or sucrose integrated with poly (vinyl alcohol) matrices. They have also demonstrated that dissolution and transiency of polymer composites could be retarded or enhanced by addition of gelatin or sucrose at different ratios, respectively.

Researchers are experimenting with a blend of programmable biodegradable and transient insulating polymer films. They have found that, by adding gelatin to the mix, dissolution can be slowed, while addition of sucrose speeds up the rate of transiency.

Using these special polymers, researchers were able to build and test an antenna that was capable of sending data and then completely dissolving when a trigger was activated. One constant in this experimentation with different composite structures is that the material maintains the appropriate physical properties to function as a substrate for electronics.

Scientists have tested several biodegradable materials including DNA, proteins and metals for making transient electronics. Tiny electronic sensors and devices that can be implanted in the body and then dissolve almost without a trace are getting closer to reality.

Efforts are on to develop a transient memory resistor with dissolvable components. This electronic component, also called a memristor, is a new type of resistor that regulates the flow of electric current and